

Volume III · Chapter 13 — Vector Databases & Knowledge Storage · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-13-context.md` for the AI interaction log.

Prerequisites: Ch 12.

Learning outcomes — the student can: explain embeddings storage and ANN search; use Chroma/FAISS (and know Weaviate/Pinecone); design metadata & filtering; reason about where knowledge lives in hardware (RAM/NVMe) and how it scales.

Topics (theory) — 8 h:

#	Topic	h
13.1	Vector embeddings & similarity metrics (cosine, dot, L2)	1
13.2	ANN indexes: HNSW, IVF; recall vs. speed	2
13.3	Vector DBs compared: Chroma, Weaviate, Pinecone, FAISS	2
13.4	Metadata, filtering & hybrid (keyword+vector) search	1
13.5	Where knowledge lives: RAM vs. NVMe; persistence; scaling	2

Datasets/tools: Chroma, FAISS, Weaviate (optional); embeddings model. **Assessment:** vector-store lab (**50%**); benchmark report (**30%**); quiz (**20%**). **Key decisions:** which vector DB; index type vs. recall/latency; RAM vs. disk; managed vs. local. **References:** plan §13 → *RAG, vector databases & local inference*. **Hours:** Theory **8** + Lab **12** = **20**.